Intensity Derivation

Double-Slit Experiment:

⇒ Consider 2 waves

$$y_1 = y_0 \sin(\omega t)$$

$$y_2 = y_0 \sin(\omega t + \Delta\phi)$$

$$\Delta\phi = \frac{2\pi}{\lambda}(d \sin\Delta\theta)$$

$$\frac{I_R}{I_0} = \left[\frac{y_{R,m}}{y_0}\right]^2$$

$$y_1 + y_2 = y_R$$

$$= y_0 \left[\sin(\omega t) + \sin(\omega t + \Delta\phi) \right]$$

$$\sin a + \sin b = 2 \cos\left[\frac{a-b}{2}\right] \sin\left[\frac{a+b}{2}\right]$$

$$\therefore y_1 + y_2 = \underbrace{2 y_0 \cos\left(\frac{\Delta\phi}{2}\right)}_{y_{R,m}} \cos\left[\omega t + \frac{\Delta\phi}{2}\right]$$

$$y_{R,m} = 2 y_0 \cos\left(\frac{\Delta\phi}{2}\right)$$

$$\frac{I_R}{I_0} = \left[\frac{y_{R,m}}{y_0}\right]^2$$

$$= \left[2\cos\left(\frac{\Delta\phi}{2}\right)\right]^2$$

$$\boxed{\frac{I}{R} = 4\frac{I}{0}\cos^2\left(\frac{\Delta\phi}{2}\right)}$$